

Tool or Social Actor? How AI Perception and Interaction Modality Influence Perceived Credibility of Communicative AI Over Time

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Abstract

People increasingly interact with conversational agents based on artificial intelligence (AI) for a variety of tasks. While these communicative AI (comAI) systems often produce responses that are natural and engaging, the information they provide is not always correct and unbiased. This makes it essential to understand how users perceive the credibility of such systems. Drawing on data from four waves of a longitudinal survey ($N_{T_1} = 618$, $N_{T_4} = 485$), we investigated how active users' perceptions of comAI as a tool or social actor and their primary interaction modality (text vs. voice) influence the credibility perceptions of comAI, both cross-sectionally and over time. We found that, within waves, social presence (T1), enjoyment, and perceived intelligence mediated the impact of tool-social actor perception but not modality on perceived credibility. A random-intercept cross-lagged panel model (RI-CLPM) did not support mediation effects at the within-person level over time. Our study contributes to the field of Human-Machine Communication with novel longitudinal insights into the psychological mechanisms that shape users' credibility judgments of comAI.

Keywords: social presence, enjoyment, perceived intelligence, large language model, longitudinal study

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Introduction

Driven by advancements in artificial intelligence (AI) technologies, people increasingly interact with language-based AI systems, such as ChatGPT, Claude, Alexa, and Siri, in their everyday lives. Take ChatGPT: as one of the fastest growing apps ever, almost 10% of all adults worldwide used it in July 2025 (Chatterji et al., 2025; Vaghasiya, 2025). Language-based AI systems are used for a wide range of tasks. Users often seek assistance with text creation, data analysis, language translation, and travel planning; look for entertainment and social support; and, increasingly, search for all types of information (e.g., politics, health, products) (Anderl, Klein, Ehrhardt, & Utz, 2024; Rowlands, 2025).

Language-based AI systems, such as ChatGPT, Claude, Alexa, and Siri, can be subsumed under the umbrella term “communicative AI” (comAI), referring to automated systems “[...] designed for the central purpose of communication [...]” (Hepp et al., 2023, p. 48). ComAI is viewed as a communicator, not as a mere tool or channel (Guzman, 2018).

A recent survey of 1,000 German participants revealed that, particularly when searching for complex information, users increasingly resort to comAI, such as ChatGPT and DeepSeek (CLANEO, 2025). However, while comAI systems, especially those based on large language models, promise natural and engaging conversations, they also pose significant risks, particularly due to their continued tendency to hallucinate, for example, by fabricating links and citing syndicated articles (Jaźwińska & Chandrasekar, 2025; Kalai et al., 2025). A recent investigation by the European Broadcasting Union (EBU) across 18 countries found that 45% of LLM-based chatbot responses to questions about news and current affairs contained at least one major issue related to accuracy, sourcing, fact-opinion distinction, or contextualization (Fletcher et al., 2025). However, what matters most for users when searching for information online is getting trustworthy results (CLANEO, 2025). This underscores the need to investigate how users form credibility perceptions of comAI to ensure responsible and informed use.

A recent meta-analysis showed that both characteristics of the recipient, such as topic involvement or emotions, as well as features of the channel, such as modality and interactivity, influence credibility perceptions (Ou & Ho, 2024). These findings are corroborated by several experimental studies, which demonstrate that the presentation mode matters for credibility assessment of comAI: For instance, when information is presented by a voice assistant, perceived credibility is higher than when the same information is presented as a textual search snippet from a classic search engine (Gaiser & Utz, 2024). Building on this, Anderl, Klein, Sarigül, et al. (2024) found that participants who received (inaccurate) pieces of information as static text (i.e., formatted like a Wikipedia article, comparable to a tool used to get information), perceived them as least credible, while participants who received the same information in a conversational form (i.e., via an LLM-based chatbot or a voice assistant) assigned higher credibility ratings (i.e., had more trouble discerning inaccurate information). Similarly, two studies compared searching information via the search engine Google and an LLM-based chatbot and found that users show overreliance on (inaccurate) information provided by LLM-based chatbots (Spatharioti et al., 2025; R. Xu et al., 2023). While previous research on credibility perceptions of comAI has focused on comparing different applications or presentation modes in experiments using vignettes or real interactions (Anderl, Klein, Sarigül, et al., 2024; Huschens et al., 2023), the current

study focused on the within-person processes taking place when people interact with such systems. We thus examined participants' subjective perceptions of comAI (hereafter: AI perception; functional as a tool vs. relational as a social actor) and usage modality (text vs. voice) when using real-world comAI systems, as well as the effects of these variables on credibility perceptions over time.

Previous research has provided some cross-sectional insights into the psychological processes underlying similar relationships. For example, Anderl, Klein, Sarigül, et al. (2024) showed that social presence mediates the effect of presentation mode (static text vs. conversational agent) on perceived information credibility. Still, other cognitive and socioemotional processes, such as perceived intelligence and enjoyment, might also play a role in the relationships between tool-social actor perception, modality, and credibility assessment. In the current study, we aimed to replicate the findings of prior experimental research (Anderl, Klein, Sarigül, et al., 2024) by examining whether social presence, but also users' enjoyment and perceived intelligence of the comAI system, take on mediating roles in the effect of tool-social actor perception and modality on credibility judgments among active comAI users in the field. Moreover, we sought to investigate how these mechanisms develop on a within-person level over time.

From a social-constructionist perspective, users' understanding of comAI (as tools or social actors) is not fixed but constructed through interaction and context (Berger & Luckmann, 2023). We argue that perceiving AI as a social actor can activate an interpretive schema that influences subsequent experiences, increasing perceptions like social presence, enjoyment, and intelligence. In this study, we focus on changes in subjective perceptions over time, distinguishing them from behavioral use patterns (i.e., interaction modality), which may reflect situational constraints rather than underlying preferences.

Our goal was to answer the following overarching research question: How do users' subjective perception (tool vs. social actor) and primary interaction modality (text vs. voice) of their favorite comAI affect credibility perceptions (over time)? We analyzed longitudinal data from a panel survey comprising four waves ($N_{T1} = 618$, $N_{T4} = 485$). To answer our research question cross-sectionally, we analyzed the data within each wave. Moreover, we wanted to isolate the theoretically interesting longitudinal within-person effects. To account for stable, trait-like between-person differences, we estimated Random-Intercept Cross-Lagged Panel Models (RI-CLPM; Hamaker et al., 2015; Little, 2024).

Prior Research and Theoretical Background

The Impact of Tool-Social Actor Perception on Credibility

The Computers Are Social Actors (CASA) paradigm suggests that people treat computers as if they were actual (human) social beings by transferring social scripts learned in human-human interactions to human-machine interactions (Nass & Moon, 2000; Reeves & Nass, 1996). This aligns with what S. Xu and Li (2024) refer to as relational use, where users see comAI systems as *social actors*, with whom one can interact and build synthetic relationships (e.g., chatting with an AI for emotional support after a stressful day) (Starke et al., 2024). Relational usage is reinforced by social design cues (Nass & Moon, 2000).

However, research indicates that users may perceive comAI not solely as social actors, but also as *tools* (Marakas et al., 2000; S. Xu & Li, 2024). From this more functional perspective, comAI systems primarily support task performance and provide assistance (e.g., checking the weather forecast). Empirical data suggests that comAI is still predominantly used for functional purposes, such as information search, creativity, and productivity support (Anderl, Klein, Ehrhardt, & Utz, 2024), while comparatively fewer users rely on them for affective conversation, such as sharing feelings, discussing personal experiences, or seeking empathetic responses (Phang et al., 2025).

Prior research suggests that credibility perceptions are dependent on individuals' personal characteristics, such as demographic features, topic involvement, and emotions (Ou & Ho, 2024). Similarly, whether users perceive a comAI system rather as a tool or a social actor may be a receiver-based factor influencing credibility assessment.

We define credibility as "judgments made by a perceiver concerning the believability of a communicator" (O'Keefe, 2016, p. 12), here referring to users' judgments about comAI as an information source. Research commonly distinguishes two key dimensions of (source) credibility (Hovland et al., 1953; McCroskey & Young, 1981; Ohanian, 1990): expertise, the extent to which a source is seen as knowledgeable, and trustworthiness, the extent to which a source is seen as honest. This study focuses on trustworthiness (Flanagin & Metzger, 2000).

One of the few longitudinal studies on the functional and relational use of voice assistants found that when users adopt a relational perspective (i.e., see their voice assistant as a social actor), they are more willing to disclose personal matters (S. Xu & Li, 2024). From a theoretical standpoint, this aligns with the CASA paradigm: When comAI is perceived as a social actor, users attribute agency, intentionality, and competence, thereby enhancing credibility judgments. We thus focus on the directional effect of AI perception on credibility, while acknowledging a potential bi-directional relationship: Higher perceived credibility (reflecting trustworthiness) may also enhance perceived agency and, in turn, lead users to construe comAI as a social actor rather than a tool.

Consistent with CASA, predominantly experimental evidence shows that increasing the perceived human-likeness or agency of comAI elicits a range of cognitive, affective, and behavioral social responses (Blut et al., 2021; Klein, 2025; Rapp et al., 2021), including higher levels of trust and perceived credibility (Anderl, Klein, Sarigül, et al., 2024; K. Xu et al., 2023). Building on this, we expected that perceiving a comAI as a social actor, rather than a mere tool, would enhance its perceived credibility.

H1: Perceived credibility is higher when people perceive comAI systems as social actors (vs. tools).

Several psychological mechanisms may underlie the effect of AI perception on perceived credibility. The first mechanism is social presence, defined as "a psychological state in which virtual (para-authentic or artificial) social actors are experienced as actual social actors in either sensory or nonsensory ways" (K. M. Lee, 2004, p. 45). In contrast to earlier definitions developed for mediated communication contexts (Biocca et al., 2003; Short et al., 1976), K. M. Lee's definition focuses on people's experience when interacting with

artificial entities that exhibit human-like qualities, such as comAI. Similarly, Cummings and Wertz (2023) define social presence as “the perceptual salience of another social actor” (p. 11), aligning the definition with commonly used operationalizations of social presence. Social presence is likely to be stronger when users treat the AI as a social entity. Moreover, social presence is an established mediator of the relationship between various visual and verbal social cues and social responses (Lombard & Xu, 2021; Van der Goot & Etzrodt, 2023).

Second, relational perception (i.e., perceiving AI as a social actor) can increase enjoyment, which is understood as the extent to which system use is inherently pleasurable independent of task-related outcomes (Davis et al., 1992). While Anderl, Klein, Sarigül, et al. (2024) found that users enjoyed reading information from Wikipedia (used as an information search tool) more than interacting with an LLM-based chatbot, other research suggests the opposite: R. Xu et al. (2023) experimentally compared differences in user experience between information search using ChatGPT and Google; participants using ChatGPT enjoyed the search task significantly more than Google users. Similarly, participants in another study comparing LLM-based and traditional search also found the LLM-based search process more satisfying (Spatharioti et al., 2025). Experimental studies comparing machine-like and human-like chatbots yield similar results. For instance, a Wizard of Oz experiment in which participants interacted with a movie recommendation chatbot exhibiting either high or low levels of self-disclosure and reciprocity (used to cue human-likeness), respectively, indicated that perceived enjoyment was higher for the human-like chatbot (S. Lee & Choi, 2017).

Third, perceiving AI as a social actor can also enhance perceived intelligence, that is, the perceived capacity of a system to engage in intelligent behavior, such as solving problems, learning, and reasoning (Bartneck et al., 2009). Two meta-analyses on AI-based agents and social robots have previously linked (perceived) anthropomorphism to perceived intelligence (Blut et al., 2021; Roesler et al., 2021). We argued that when a comAI system is perceived as a social actor, users attribute greater agency, including intentionality and self-reactiveness (Bandura, 2001). In contrast, when perceived as a tool, it may be viewed as instrumentally useful but not necessarily as intelligent in an agentic sense. We thus proposed:

H2: Social presence (a), enjoyment (b), and perceived intelligence (c) are higher when people perceive comAI systems as social actors (vs. tools).

Prior research has shown that social presence is associated with increased credibility perceptions and reduced fact-checking (Anderl, Klein, Sarigül, et al., 2024; Jun et al., 2017), and that enjoyment similarly correlates with credibility perceptions (Anderl, Klein, Sarigül, et al., 2024). These relationships may stem from a shift from systematic toward more heuristic processing, facilitated by social presence and enjoyment (Mikkilineni & Lee, 2025). Interactions that feel socially present or enjoyable may be more fluent to process, reducing cognitive effort and the motivation for critical evaluation (Breves, 2023; Stadler et al., 2024). As a result, users may process interactions as if they were engaging with a socially present source, making it more credible (Sundar, 2008). Enjoyment may increase credibility through affective misattribution: Content that “feels good” is more likely to be judged

as credible because the ease and positivity of the interaction are implicitly attributed to the source.

The third potential mediator, perceived intelligence, is conceptually related to the expertise dimension of established source credibility frameworks (McCroskey & Young, 1981; Ohanian, 1990; O’Keefe, 2016). Our concept of credibility primarily reflects trustworthiness rather than competence. We therefore argued that perceived intelligence may function as a heuristic performance cue: When a source appears intelligent, users may infer that it is also reliable, leading to higher credibility judgments (Ou & Ho, 2024; Sundar, 2008).

Taken together, these mechanisms may help explain why perceiving a comAI system as a social actor rather than a functional tool may be associated with higher credibility perceptions. We thus hypothesized:

H3: Social presence (a), enjoyment (b), and perceived intelligence (c) mediate the effect of AI perception (tool vs. social actor) on perceived credibility.

The Impact of Primary Interaction Modality on Credibility Perceptions

Modality refers to features of the channel, that is, to how specific information is presented to the user (e.g., via text or audio). In this study, primary interaction modality refers to the mode people use most frequently, rather than their preferred or desired mode. Channel features have also been shown to affect perceptions of credibility (Ou & Ho, 2024). In Sundar’s Modality-Agency-Interactivity-Navigability model (2008), modality is one of the four main affordances said to influence users’ psychological reactions to media technologies. In this context, the realism heuristic refers to the rule of thumb that information presented in text form leaves more room for interpretation and ambiguity than the same information presented more realistically (e.g., via audio or audiovisual means). Audio or audiovisual formats may be perceived as more authentic, leading to higher credibility ratings than textual presentation (“hearing is believing”) (Sundar, 2008). At the same time, unlike answers from voice assistants, textual information can be re-read and scrutinized, which may increase the likelihood of detecting inaccuracies and thus decrease perceived credibility. However, written information is often processed more fluently than spoken information (Schwede et al., 2022). This increased processing fluency (i.e., “the subjective experience of ease or difficulty associated with completing a mental task”; Oppenheimer, 2008, p. 237) can indirectly enhance credibility perceptions (Anderl, Klein, Sarigül, et al., 2024).

We build on the findings of Gaiser and Utz (2024), who showed that credibility is higher when information is received from a voice assistant (vs. formatted like a Google search snippet), and Anderl, Klein, Sarigül, et al. (2024), who replicated this finding for identical information from voice-based agents like Alexa vs. textual Wikipedia snippets. In the current study, we thus assumed that people perceive voice-based comAI as more credible than text-based comAI.

H4: Perceived credibility is higher when people interact with comAI systems via voice (vs. text).

Prior research has found associations between modality and enjoyment, for instance, that it is more enjoyable to talk (vs. write) to a comAI system (Rzepka et al., 2022) and that speech interactions in service lead to more flow-like and overall more positive experiences compared to text interactions (Zierau et al., 2023). Some studies conclude that it depends: In an e-commerce context, talking is favored for hedonic products, such as action movies, while writing is favored for utilitarian products, such as documentaries (Schindler et al., 2024). Other studies found reverse effects, that is, text-only was perceived as more human-like than voice (Ischen et al., 2022), or null effects (Anderl, Klein, Sarigül, et al., 2024). As prior research did not find clear evidence that receiving information from a text-based (vs. voice-based) comAI leads to higher social presence, enjoyment, and perceived intelligence (Anderl, Klein, Sarigül, et al., 2024), we formulated the following research questions:

RQ1: Does modality (text vs. voice) influence social presence (a), perceived enjoyment (b), and perceived intelligence (c)?

RQ2: Do social presence (a), perceived enjoyment (b), and perceived intelligence (c) mediate the effect of modality (text vs. voice) on perceived credibility?

Longitudinal Hypotheses and Research Questions

Prior research on credibility perceptions of comAI has been mostly experimental. Researchers manipulated various system characteristics (e.g., presentation mode or modality) and examined their isolated or interaction effects on perceived credibility (Anderl, Klein, Sarigül, et al., 2024; Huschens et al., 2023). Such experimental designs provide highly controlled environments and allow causal statements, at least for the paths that depart from the manipulated experimental factors. They are less informative, though, regarding the directionality of relationships between how people *perceive* a system, the potential mediators, and their credibility assessment. Therefore, reciprocal effects between tool-social actor perception, the mediators, and credibility are theoretically plausible and cannot be ruled out. However, our primary interest lies in the extent to which tool-social actor perception influences credibility over time.

Building on this, we investigate the above relationships with a longitudinal approach, which offers a more nuanced understanding of cause-and-effect relationships (Cole & Maxwell, 2003; Jose, 2016). With this approach, we can separate meaningful within-person effects from between-person differences (i.e., selection effects), allowing us to examine how changes in perceptions within individuals—as more tool-like or more social actor-like—relate to changes in the potential mediators and credibility. In line with a social constructionist perspective, we conceptualized tool-actor perception as a fundamental dimension that shapes related perceptions of the system.

Prior work has found differences in credibility perception between systems such as ChatGPT and web search (Anderl, Klein, Sarigül, et al., 2024; Huschens et al., 2023). To extend this line of research, we investigate whether similar differences emerge depending on whether individuals construe their favorite AI more as a tool (comparable to web search) or as a social actor. Based on the theoretical considerations and empirical evidence on the relation between AI perception (tool vs. social actor), the potential mediators, and perceived credibility, we proposed the following hypotheses at the within-person level:

H5: Within persons, perceiving an AI as a social actor (vs. a tool) at time t is associated with higher perceived credibility at time $t+1$.

H6: Within persons, perceiving an AI as a social actor (vs. a tool) at time t is associated with higher social presence (a), enjoyment (b), and perceived intelligence (c) at time $t+1$.

We expected the effect described in H5 to operate through heightened perceptions of social presence, enjoyment, and perceived intelligence.

H7: Within persons, social presence (a), enjoyment (b), and perceived intelligence (c) at time $t+1$ mediate the effect of AI perception (tool vs. social actor) at time t on perceived credibility at time $t+2$.

As prior research has produced inconclusive results regarding the effect of modality (text- vs. voice-based conversational agent) on credibility perceptions (Anderl, Klein, Sarigül, et al., 2024), we formulated the following research questions:

RQ3: Within persons, is the primary interaction modality (text vs. voice) at time t associated with higher perceived credibility at time $t+1$?

RQ4: Within persons, is the primary interaction modality (text vs. voice) at time t associated with higher social presence (a), perceived enjoyment (b), and perceived intelligence (c) at time $t+1$?

RQ5: Within persons, do social presence (a), perceived enjoyment (b), and perceived intelligence (c) at time $t+1$ mediate the effect of primary interaction modality (text vs. voice) at time t on perceived credibility at time $t+2$?

Method

Participants and Procedure

This study was based on Waves 2–5 (hereafter time points T1–4) of a large longitudinal survey spanning six waves over the course of 10 months, with one wave conducted every 2 months. A detailed description of the complete longitudinal dataset is provided in a separate data descriptor paper (Henestrosa et al., 2026). The project was approved by the ethics committee of the Leibniz-Institut für Wissensmedien (LEK 2024/031). $N = 1,008$ US-based (as new LLMs are typically first published in the US) users of comAI systems and people open to using them had been invited to Wave 1 in September 2024; data collection for Wave 5 took place in May 2025.¹ Data were collected using a Qualtrics questionnaire distributed via the online sampling platform Prolific. Due to varying questionnaire lengths, participants received between £2.74 (T1) and £3.50 (T4) for their participation.

1. Following Wave 3, participants from Wave 1 who had not participated in Waves 2 and/or 3 were recontacted to take part in Waves 4–6. This decision aimed to increase statistical power for analyses that did not require complete data across all waves.

TABLE 1
Sample Characteristics Per Wave

	T1	T2	T3	T4
<i>N</i>	618	544	504	485
Attrition rate (%)	—	11.97	7.35	3.77
Female (%)	48.87	50.92	51.19	52.37
Age 18–29 (%)	13.92	13.60	11.11	11.13
Age 30–39 (%)	40.13	38.79	39.88	38.24
Age 40–49 (%)	25.40	27.21	26.59	26.80
Age 50–65 (%)	16.99	17.46	18.85	20.62
Age 65+ (%)	3.56	2.94	3.57	3.30
% social actor (vs. tool)	12.78	13.60	13.10	13.61
% voice (vs. text)	9.39	11.58	11.11	8.25

Note. T1–T4 refer to Waves 2–5, respectively.

The current study was preregistered: <https://doi.org/10.17605/OSF.IO/F3JYC>. Wave 1 of the larger project served as a pilot study; see our OSF repository² for results: <https://doi.org/10.17605/OSF.IO/2A7MY>. Data collection for Wave 2 (the first time point in our current study) was completed shortly after planning for this study was finished. However, we did not receive access to the Wave 2 data until after submitting the preregistration.

Participants had to be at least 18 years old and consent to participation and to the use of their anonymized data for scientific purposes. Additionally, participants for the current study had to be at least one-time users of language-based comAI systems. Table 1 shows the sample characteristics, including sample sizes and attrition rates for the four included waves.

Measures

Independent Variables

Before asking participants about their tool-social actor perception and favorite usage modality, we asked them what their favorite AI system is. If participants did not have a favorite AI system, we asked them about the system they used most often. In addition, participants were reminded that by AI, we meant language-based AI systems. Answer options included ChatGPT, Microsoft Copilot (formerly Bing Chat), Claude, Google AI incl. Google Gemini (formerly Bard), Alexa, Siri/Apple Intelligence, Samsung Galaxy AI, Twitter/X Grok, Meta AI, and other (open response). Most participants named ChatGPT as their favorite system (67% in W2, 65% in W3, 67% in W4, and 66% in W5). This was followed by Google AI/Gemini (around 10%) and Copilot (around 7%), while Alexa and Siri were mentioned less frequently (around 5% each).

2. The OSF repository also contains an overview of the deviations from the preregistration, the study measures, and additional results.

AI perception was assessed using the self-developed question “How do you usually perceive [favorite AI system] while using it?” Answer options were “more like a tool (an instrument or machine you use to get things done)” or “more like a social actor (some form of social being you can interact with).”

Modality was assessed using the self-developed question “How do you interact with [favorite AI system] most often?” Participants could select “mostly via voice,” “mix of voice and text,” or “mostly via text.” As preregistered, we dichotomized the variable by excluding the option “mix of voice and text” and retaining only participants who indicated interacting “mostly via voice” or “mostly via text.” Across waves, most participants (>55%) in the text group identified ChatGPT as their favorite AI system, whereas in the voice group, most participants (>85%) reported Alexa or Apple AI/Siri as their favorite. The dichotomization was motivated by the small size of the mix group ($M = 14.4\%$), and pilot findings showing that this group reported the highest levels of credibility, social presence, enjoyment, and perceived intelligence—contrary to our expectations that these would be highest for voice.

Dependent Variable and Potential Mediators

Perceived credibility of favorite AI system was measured using five items adapted from Flanagan and Metzger (2000): [Favorite AI system is] [unbelievable—believable, inaccurate—accurate, untrustworthy—trustworthy, biased—unbiased, not credible—credible]. These items were assessed on 5-point semantic differential scales. Cronbach’s α ranged from .88 (T1) to .93 (T4).

We measured social presence while using one’s favorite AI system using five items adapted from Gefen and Straub (2004): There is a sense of [human contact, personalness, sociability, human warmth, sensitivity] in the interaction. Items were assessed on Likert-type rating scales from 1 (*strongly disagree*) to 7 (*strongly agree*). Cronbach’s α ranged from .96 (T1) to .97 (T4).

To measure enjoyment, three items adapted from Venkatesh (2000) were used: I find using [favorite AI system] to be enjoyable, pleasant. I have fun using [favorite AI system]. Enjoyment was assessed using Likert-type rating scales from 1 (*strongly disagree*) to 7 (*strongly agree*). Cronbach’s α ranged from .93 (T1) to .95 (T4).

Perceived intelligence was assessed with five items adapted from Bartneck et al. (2009)’s Godspeed questionnaire, developed explicitly in the context of human-machine interaction: [Favorite AI system is] [foolish—sensible, incompetent—competent, unintelligent—intelligent, irresponsible—responsible, stupid—knowledgeable] using 5-point semantic differential scales. Cronbach’s α ranged from .90 (T1) to .92 (T4).

Descriptive statistics and bivariate correlations among the model variables at each time point are presented in Table 2. The constructs were relatively stable across time points (gray-shaded cells, $.68 < r < .82$, $p < .001$). Perceived credibility, social presence, enjoyment, and perceived intelligence were significantly positively correlated. Particularly credibility and perceived intelligence were highly correlated within waves ($.72 < r < .82$). Therefore, we assessed discriminant validity by conducting exploratory and confirmatory factor analyses. The standardized factor loadings indicated that the items loaded on their respective factors, and correlations between latent variables fell within an acceptable range (see OSF).

TABLE 2
Descriptive Statistics and Bivariate Correlations Among Model Variables

Var	<i>M (SD)</i>	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.
1. Cre1	3.97 (0.68)	—															
2. Cre2	3.96 (0.72)	.74	—														
3. Cre3	3.92 (0.77)	.73	.77	—													
4. Cre4	3.92 (0.77)	.72	.76	.80	—												
5. SoP1	3.43 (1.78)	.37	.37	.39	.33	—											
6. SoP2	3.40 (1.77)	.33	.38	.39	.36	.79	—										
7. SoP3	3.39 (1.81)	.30	.35	.41	.37	.77	.78	—									
8. SoP4	3.52 (1.85)	.33	.34	.40	.38	.77	.77	.79	—								
9. Enj1	5.47 (1.17)	.57	.51	.54	.55	.40	.32	.38	.39	—							
10. Enj2	5.47 (1.26)	.58	.63	.60	.58	.38	.44	.41	.40	.73	—						
11. Enj3	5.46 (1.32)	.54	.57	.64	.58	.40	.39	.47	.43	.72	.79	—					
12. Enj4	5.41 (1.35)	.57	.60	.61	.63	.44	.42	.46	.48	.74	.82	.82	—				
13. Int1	4.10 (0.71)	.71	.60	.63	.67	.35	.34	.32	.37	.59	.56	.55	.59	—			
14. Int2	4.09 (0.74)	.64	.77	.67	.66	.38	.43	.40	.41	.55	.66	.59	.59	.68	—		
15. Int3	4.08 (0.75)	.68	.73	.82	.73	.39	.42	.45	.43	.56	.60	.64	.63	.69	.77	—	
16. Int4	4.09 (0.78)	.65	.71	.73	.82	.39	.42	.45	.45	.58	.58	.59	.66	.74	.74	.80	—

Note. Cre = Credibility, SoP = Social presence, Enj = Enjoyment, Int = Perceived intelligence. Pearson correlations, all $p < .05$. $N_{T1} = 618$, $N_{T2} = 544$, $N_{T3} = 504$, $N_{T4} = 485$.

Results

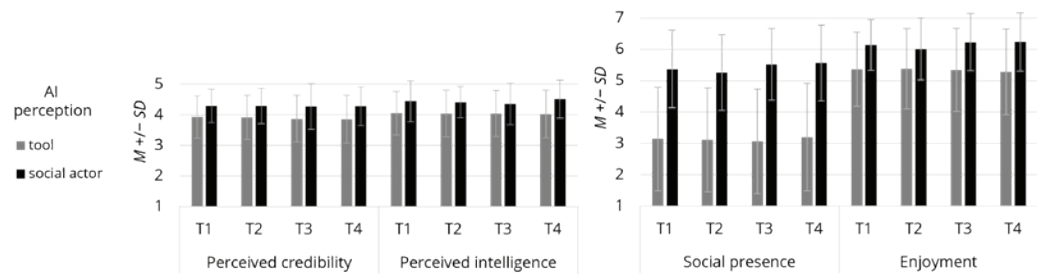
We tested our hypotheses and answered our research questions by analyzing the data cross-sectionally and longitudinally. All analyses were conducted using R version 4.4.2.

Cross-Sectional Analyses

To test H1, H2, and H4 and to answer RQ1, we first conducted a multivariate analysis of variance (MANOVA) for each predictor (i.e., AI perception, modality) with perceived credibility and the potential mediators as dependent variables at each time point. The MANOVAs were followed by one univariate ANOVA per dependent variable. To account for differences in scaling, all continuous variables were standardized ($M = 0$, $SD = 1$) prior to conducting the MANOVAs.

At all time points, using Pillai's trace, the multivariate effect of AI perception (tool vs. social actor) was significant (T1: $V = .18$, $F(4, 616) = 33.17$, $p < .001$, T2: $V = .18$, $F(4, 542) = 29.10$, $p < .001$, T3: $V = .22$, $F(4, 502) = 35.44$, $p < .001$, T4: $V = .19$, $F(4, 483) = 28.52$, $p < .001$). Additional univariate ANOVAs revealed that perceiving AI as a social

FIGURE 1
Means and Standard Deviations of Dependent Variable Perceived Credibility and Potential Mediators by AI Perception



Note. All mean differences (tool vs. social actor) $p < .001$.

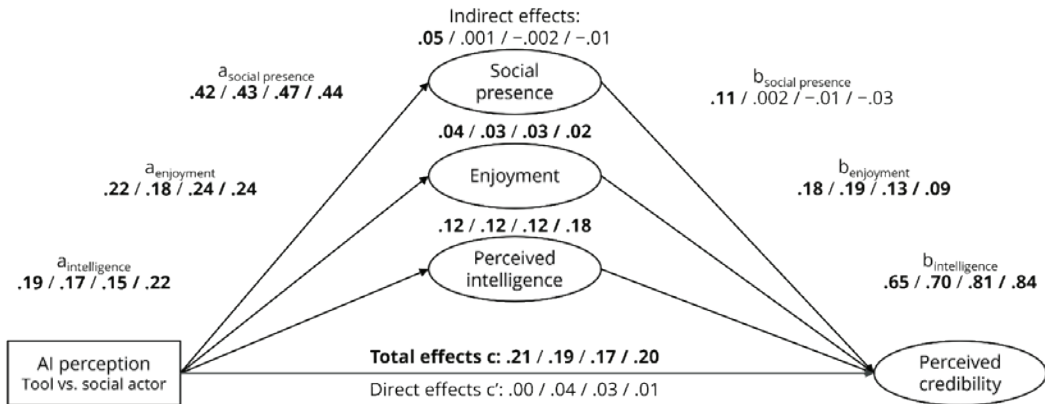
actor significantly increased perceptions of the dependent variable credibility, as well as the potential mediators, social presence, enjoyment, and intelligence (Figure 1, all $p < .001$). The first two cross-sectional hypotheses (H1, H2) could thus be confirmed.

At T1, the multivariate effect of modality (text vs. voice) was small but significant, $V = .06$, $F(4, 531) = 8.09$, $p < .001$. Subsequent univariate ANOVAs yielded significant effects on social presence, $F(1, 531) = 7.30$, $p = .007$ (text: $M = 3.38$, $SD = 1.74$, voice: $M = 2.73$, $SD = 1.53$), and enjoyment $F(1, 531) = 8.61$, $p = .003$ (text: $M = 5.47$, $SD = 1.15$, voice: $M = 4.99$, $SD = 1.46$). However, social presence and enjoyment were higher for text (vs. voice) interaction. At T3, there was also a significant multivariate effect, $V = .02$, $F(4, 431) = 2.74$, $p = .029$ with a significant univariate effect on perceived credibility, $F(1, 431) = 4.11$, $p = .043$, which went in the predicted direction (text: $M = 3.86$, $SD = 0.78$, voice: $M = 4.09$, $SD = 0.80$). There were no multivariate effects at T2, $V = .01$, $F(4, 462) = 1.00$, $p = .405$, and T4, $V = .02$, $F(4, 410) = 2.19$, $p = .069$. Therefore, H4 had to be largely rejected. Except for the (reverse) effects on social presence and enjoyment at T1, no significant effects emerged on the potential mediators (RQ1). The full results of the univariate ANOVAs with AI perception and modality as predictors are available on OSF.

To test H3 and answer RQ2, we had preregistered a parallel mediator model for each predictor using the lavaan package (Rosseel et al., 2021). As preregistered, the models included all mediators for which we found significant effects of AI perception and modality when investigating H2 and RQ1, and only the significant predictors of perceived credibility. As there was no wave in which we found significant modality effects on both perceived credibility and at least one of the potential mediators, we estimated only one parallel mediator model with AI perception (tool vs. social actor) as the independent variable, all potential mediators, and perceived credibility as the dependent variable (Figure 2).

At T1, we found a full mediation (a significant total effect and a nonsignificant direct effect), with all indirect effects via the mediators being significant. At T2, T3, and T4, the results also suggested full mediation, with significant indirect effects via enjoyment and

FIGURE 2
Results of Parallel Mediator Models Within Time Points



Note. Standardized coefficients (T1 / T2 / T3 / T4). Coefficients in bold $p < .05$. T1: $N = 618$, CFI = .977, RMSEA = .051, SRMR = .035. T2: $N = 544$, CFI = .976, RMSEA = .054, SRMR = .035. T3: $N = 504$, CFI = .973, RMSEA = .060, SRMR = .036. T4: $N = 485$, CFI = .976, RMSEA = .059, SRMR = .033.

perceived intelligence, not via social presence.³ Our cross-sectional mediation hypothesis for AI perception (tool vs. social actor) (H3) could thus be largely confirmed.

Longitudinal Analyses

To test H5–H7 and to answer RQ3–RQ5, we preregistered to estimate one random-intercept cross-lagged panel model (RI-CLPM) (Hamaker et al., 2015) using the lavaan package (Rosseel et al., 2021). The RI-CLPM approach enables researchers to simultaneously test the lagged relationship between the predictors (i.e., AI perception and modality) via the three potential mediators on perceived credibility while controlling for prior values of perceived credibility and the potential mediator variables. By including random intercepts, within-person effects (intraindividual changes over time) can be separated from stable between-person effects (interindividual differences) (Hamaker et al., 2015). To test H5 and H6 and to answer RQ3 and RQ4, we planned to examine the cross-lagged effects of AI perception or modality on the respective dependent variables. To test H7 and to answer RQ5, we planned to analyze total, direct, and indirect effects (Little, 2024; Newsom, 2015).

Model fit was assessed using established fit indices and cutoff criteria. A robust CFI (Comparative Fit Index) above .95, a robust RMSEA (Root Mean Square Error of Approximation) below .06, and an SRMR (Standardized Root Mean Square Residual) below .08 indicated good model fit (Hu & Bentler, 1999). Before running the main longitudinal analyses, we conducted attrition analyses and measurement invariance testing.

3. As perceived intelligence and credibility were highly correlated, we additionally estimated mediation models excluding perceived intelligence to explore potential suppressor effects. Compared to the preregistered model, significant indirect effects via social presence and enjoyment emerged in the first three waves but were not observed in the final wave. Full results are available on OSF.

Attrition Analysis

Regarding item nonresponse within time points, we did not have missing values in the model variables. Regarding wave nonresponse, we examined whether participants who dropped out before the last wave (T4) differed from those who participated in all waves (T1–T4) on the model variables and on age, gender, and education. We did not find significant differences regarding gender, $\chi^2(3, 618) = 2.74, p = .433$, and education, $t(446.92) = -1.78, p = .075$. However, participants who completed all waves were older at the first time point than those who dropped out before the last time point ($M_{\text{Dropout}} = 36.40, M_{\text{All-Waves}} = 40.82, t(452.81) = 3.67, p < .001$). Age was positively related to perceived credibility ($p < .001$), enjoyment ($p = .001$), perceived intelligence ($p = .003$), but not to social presence ($p = .524$). Education was negatively related to perceived credibility ($p < .001$), social presence ($p = .042$), and perceived intelligence ($p < .001$) but not to enjoyment ($p = .261$).

In addition, we tested whether participants who completed one, two, three, or all waves differed on the dependent variables using an ANOVA with subsequent post-hoc tests. (Only) 1 out of 20 differences was significant: One-wave versus all-wave participants differed significantly in enjoyment, with 4-wave participants scoring higher ($\Delta M = 0.37, p = .006$). As this was the only significant difference, we assumed that missing values were missing at random (MAR) (Enders, 2010). We thus estimated the RI-CLPM using the Full Information Maximum Likelihood (FIML) estimator, allowing us to also use data from participants with missing values in some waves (Newsom, 2015).

Measurement Invariance

A necessary condition to conduct longitudinal structural equation models is that the model variables are invariant across time (Newsom, 2015). Prior to model estimation, we tested for invariance of the measurement model across time points using the lavaan package (Rosseel et al., 2021). The measurement model included the model variables perceived credibility, social presence, enjoyment, and perceived intelligence. The results indicated strong (scalar) measurement invariance across time (Table 3).

TABLE 3
Measurement Invariance Testing Results

	Fit indices								Chi-square difference test		
Model	n _{par}	χ ²	df	Robust CFI	Robust RMSEA	SRMR	AIC	BIC	Δχ ²	Δdf	p
Configural invariance	240	1475.16	516	.975	.059	.034	84141.72	85503.40	—	—	—
Metric invariance	198	1510.36	558	.975	.056	.037	84092.92	85216.31	35.21	42	.762
Scalar invariance	156	1552.82	600	.975	.054	.038	84051.38	84936.48	42.46	42	.451

Note. Measurement model consisting of social presence, enjoyment, perceived intelligence, and perceived credibility. n_{par} = number of parameters estimated; χ^2 = chi-square statistic; CFI = comparative fit index, RMSEA = root mean square error of approximation; SRMR = standardized root mean square residual; AIC = Akaike information criterion; SABIC = Bayesian information criterion.

Model Setup

We had preregistered to run one RI-CLPM including both predictors (i.e., AI perception and modality). However, the full model proved highly complex and resulted in estimation problems, specifically, negative latent variances and, thus, non-admissible solutions, indicating model instability (Kline, 2016). Model estimates were not reliable, likely due to the high model complexity relative to the available sample size (see OSF). To obtain reliable estimates, we ran separate models for each predictor using robust maximum likelihood (MLR), while retaining all paths. Estimating separate models was a reasonable approach because we did not assume that the predictors were correlated. The models were constructed with random intercepts (RI), representing the between-person levels for the model variables. We modeled all autoregressive and cross-lagged paths, which indicate potential within-person associations among the model variables, including reverse paths to assess potential bi-directional relationships (Hamaker et al., 2015). Autoregressive and cross-lagged paths were constrained to be equal across waves to improve model stability (Hamaker et al., 2015).

The model with AI perception as the predictor converged successfully and yielded reliable estimates. We estimated two versions of this model: one as preregistered (without covariates) and one that included age and education as covariates, because significant age and education differences were observed between participants who dropped out and those who completed all waves (see Attrition Analysis). All continuous variables were standardized ($M = 0$, $SD = 1$) prior to estimating the models. The relevant results for the AI perception models are reported below; full results are available on OSF.

Regarding the model with modality as predictor: The estimated model was not reliable, given negative latent variances (see OSF for details). While the longitudinal RQ3–5 could not be answered with our current data, the cross-sectional findings suggested that participants' primary interaction modality may ultimately have only limited impacts on social presence, enjoyment, perceived intelligence, and perceived credibility.

Model Fit and Parameters: AI Perception Without Covariates

Model fit of the model without covariates was acceptable (robust CFI = .998, robust RMSEA = .019, SRMR = .022). Tables 4 and 5 show the relevant results of the model without covariates.

At the between-person level, as reflected in the random intercepts, all model variables were positively associated. For example, individuals who reported higher social presence also tended to perceive their favorite comAI system as more credible.

At the within-person level, the autoregressive path for credibility was positive and significant, indicating some temporal stability within individuals for this variable. In other words, a person's deviation in credibility from their mean credibility at time t predicted their deviation in credibility at time $t+1$. More interestingly, there were also significant cross-lagged effects at the within-person level. AI perception (tool vs. social actor) had a positive cross-lagged effect on social presence, and social presence had a positive cross-lagged effect on perceived credibility. When a person's perception shifted toward viewing the AI as a social actor (as opposed to a tool) at one time point, their social presence score at the next time point tended to be higher than usual. Similarly, when a person

TABLE 4
Covariances (Within-Person) and Random Intercepts (Between-Person)

					Mediator					
					Social presence		Enjoyment		Perceived intelligence	
Path			β	p	β	p	β	p	β	p
Tool_actor	↔	Mediator								
T1					.02	.021	.01	.257	.01	.228
T2					.03	.005	.01	.410	< .001	.979
T3					.04	< .001	.01	.143	.001	.868
T4					.02	.033	.002	.718	.01	.393
Random intercept					.12	< .001	.07	< .001	.06	< .001
Mediator	↔	Credibility								
T1					.02	.439	.07	.005	.14	< .001
T2					.04	.012	.08	< .001	.12	< .001
T3					.05	.013	.08	.001	.11	< .001
T4					.03	.061	.04	.020	.09	< .001
Random intercept					.36	< .001	.56	< .001	.68	< .001
Tool_actor	↔	Credibility								
T1			-.01	.396						
T2			.002	.746						
T3			.02	.104						
T4			.01	.316						
Random intercept			.05	< .001						

Note. Tool_actor: perception of favorite AI system as tool vs. social actor. Model without covariates. Standardized coefficients β . Coefficients in bold $p < .05$.

experienced higher-than-usual social presence at one time point, they tended to report higher-than-usual credibility at the next time point. However, the indirect effect of AI perception on credibility via social presence was not significant, and neither was the total effect of AI perception on credibility.

In addition, although not relevant to testing our hypotheses, social presence exhibited cross-lagged effects on AI perception and enjoyment. Within individuals, higher social presence at one time point was associated with shifts in AI perception from tool to social actor ($\beta = .06, p = .019$) and higher enjoyment ($\beta = .08, p = .049$) at the next time point. No further significant cross-lagged paths emerged.⁴

4. Like in the cross-sectional analysis, we estimated an RI-CLPM excluding perceived intelligence to explore potential suppressor effects. However, the overall pattern of results remained unchanged (OSF).

TABLE 5
Auto-Regressive and Cross-Lagged Estimates (Within-Person)

				Mediator						
				Social presence		Enjoyment		Perceived intelligence		
Path		β	p	β	p	β	p	β	p	
Auto-regressive										
Tool_actor		.09	.252							
Mediator				.08	.188	-.01	.863	.06	.357	
Credibility		.13	.031							
Cross-lagged										
Tool_actor	→	Mediator			.15	.043	-.05	.528	.07	.417
Mediator	→	Credibility			.11	.012	.01	.877	-.06	.207
Tool_actor	→	Credibility	-.05	.610						
Credibility	→	Mediator			.08	.098	.07	.194	.06	.284
Mediator	→	Tool_actor			.06	.019	.004	.828	-.03	.171
Credibility	→	Tool_actor	-.003	.884						
Indirect Tool_actor	→	Mediator → Credibility			.02	.106	.00	.885	-.004	.498
Total Tool_actor	→	Credibility			-.03	.710				

Note. Tool_actor: perception of favorite AI system as tool vs. social actor. Model without covariates. Standardized coefficients β . Coefficients in bold $p < .05$.

Model Fit and Parameters: AI Perception With Covariates

Model fit of the model with covariates was good (robust CFI = 1.00, robust RMSEA = .003, SRMR = .018). Tables 6 and 7 show the relevant results of the model with time-invariant covariates age and education.

The estimation yielded results similar to those in the model without covariates. We again found significant positive associations at the between-person level (random intercepts), a significant positive autoregressive path for credibility, as well as significant cross-lagged paths from AI perception to social presence (partially confirming H6) and from social presence to perceived credibility. A significant autoregressive path emerged for AI perception. In contrast to the model without covariates, the indirect effect of AI perception on credibility via social presence was marginally significant when age and education were included as control variables. However, the total effect of AI perception on credibility was again not significant. We thus rejected H5 and our longitudinal mediation hypothesis (H7).

The positive cross-lagged paths from social presence to AI perception ($\beta = .06$, $p = .041$), and enjoyment ($\beta = .09$, $p = .048$) persisted, and an additional positive cross-lagged path to perceived intelligence ($\beta = .11$, $p = .026$) emerged.

TABLE 6
Covariances (Within-Person) and Random Intercepts (Between-Person)

					Mediator					
					Social presence		Enjoyment		Perceived intelligence	
Path			β	p	β	p	β	p	β	p
Tool_actor	↔	Mediator								
T1					.02	.005	.01	.100	.01	.272
T2					.03	.002	.01	.280	.01	.510
T3					.04	< .001	.02	.072	.003	.676
T4					.02	.031	-.003	.438	.004	.523
Random intercept					.11	< .001	.06	< .001	.05	< .001
Mediator	↔	Credibility								
T1					.03	.145	.07	.003	.12	< .001
T2					.04	.030	.08	< .001	.13	< .001
T3					.04	.032	.08	.001	.13	< .001
T4					.02	.093	.04	.019	.09	< .001
Random intercept					.37	< .001	.54	< .001	.63	< .001
Tool_actor	↔	Credibility								
T1			.001	.948						
T2			< .001	.982						
T3			.01	.139						
T4			-.001	.912						
Random intercept			.06	< .001						

Note. Tool_actor: perception of favorite AI system as tool vs. social actor. Model controlled for age and education. Standardized coefficients β . Coefficients in bold $p < .05$.

Exploratory Analyses

The finding that social presence influenced enjoyment (in both models), perceived intelligence (in the model with covariates), and, of course, perceived credibility over time aligns with prior research highlighting its central role in shaping cognitive, affective, and behavioral responses to machines (Lombard & Xu, 2021). Together with the nonsignificant path from social presence to credibility in the later waves, this pattern may suggest serial mediation, in which social presence enhances enjoyment or perceived intelligence, which in turn strengthens credibility. However, adding these serial mediation paths to the RI-CLPM yielded no significant effects (full results on OSF).

TABLE 7
Auto-Regressive and Cross-Lagged Estimates (Within-Person)

					Mediator					
					Social presence		Enjoyment		Perceived intelligence	
Path			β	p	β	p	β	p	β	p
Auto-regressive										
Tool_actor			.16	.047						
Mediator					.07	.216	−.01	.866	.08	.237
Credibility			.16	.015						
Cross-lagged										
Tool_actor	→	Mediator			.18	.016	−.04	.572	.08	.354
Mediator	→	Credibility			.11	.018	.002	.977	−.06	.288
Tool_actor	→	Credibility	−.10	.269						
Credibility	→	Mediator			.04	.390	.07	.117	.08	.177
Mediator	→	Tool_actor			.06	.041	.01	.737	−.02	.363
Credibility	→	Tool_actor	−.01	.603						
Indirect Tool_actor	→	Mediator → Credibility			.02	.075	.00	.977	−.004	.517
Total Tool_actor	→	Credibility			−.08	.355				

Note. Tool_actor: perception of favorite AI system as tool vs. social actor. Model controlled for age and education. Standardized coefficients β . Coefficients in bold $p < .05$.

Discussion

With this study, we aimed to investigate how AI perception (tool vs. social actor) and interaction modality (text vs. voice) influence users' credibility perceptions. Analysis of data from four measurement points over a 6-month period provided us with both cross-sectional and longitudinal insights into the underlying psychological mechanisms.

Cross-Sectional Mechanisms of Credibility Formation

Across time points, perceiving AI as a social actor (rather than a tool) was associated with higher credibility perceptions through social presence (partially), enjoyment, and perceived intelligence. All proposed mediators played a significant role, suggesting that both socioemotional and cognitive factors influence credibility formation.

These findings conceptually replicate previous experimental research on how different presentation modes impact credibility perceptions (Anderl, Klein, Sarigül, et al., 2024; Gaiser & Utz, 2024; Spatharioti et al., 2025; R. Xu et al., 2023). In contrast to previous research, in which participants were presented with vignettes of user-AI interactions, we observed a significant mediation effect via enjoyment, suggesting that direct interaction with AI is necessary for the mediation effect to become significant. Overall, the findings support the CASA paradigm (Nass et al., 1994; Nass & Moon, 2000), indicating that individuals who perceive a comAI system as a social actor rather than a tool tend to experience greater social presence and enjoyment, and perceive the system as more intelligent when interacting with their favorite comAI. These perceptions, in turn, are positively associated with higher credibility.

From CASA to Machine-Specific Interaction Scripts

While social presence was consistently higher among people who perceived their favorite AI as a social actor (vs. a tool), the positive effect of social presence on perceived credibility observed in the parallel mediator model at T1 disappeared in later waves (in the last wave when omitting perceived intelligence as a mediator). In contrast, enjoyment and perceived intelligence mediated the relationship between AI perception and credibility in all waves. Thus, for their credibility evaluations, participants may have shifted their focus toward the enjoyment of the interaction and functional aspects, such as perceived intelligence, rather than relational aspects. The more frequently users interact with a comAI system, the more their perception of its credibility may depend on how pleasant and how intelligent they perceive comAI to be, rather than on whether it feels socially present. This shift may in part reflect the development and application of specific scripts for human-machine communication, in which a human-like interaction experience potentially becomes less central (Gambino et al., 2020).

Our findings are consistent with recent HMC studies reporting that CASA effects may attenuate in repeated interactions (politeness norms: Lazebnik et al., 2025; Qi & Zhao, 2026) or failed (conceptual) CASA replications, including regarding helping behavior toward social robots (Sharan et al., 2025) and politeness toward desktop computers (Heyseelaar, 2023). These findings suggest that, with more frequent and intense interactions, users may no longer consistently apply human social norms to their interactions with comAI. Instead, emerging evidence indicates that individuals form machine-specific interaction schemas (e.g., Piercy & Turner-Leatherman, 2025).

Taken together, our findings indicate that CASA effects may be most pronounced in early or less familiar interactions, but become less central as users develop more experience with comAI. Future research should investigate when CASA still holds (e.g., for emerging technologies; Heyseelaar, 2023), what machine-specific interaction schemas look like, and when they are applied (e.g., in more goal-directed contexts; Qi & Zhao, 2026).

Longitudinal Dynamics of Credibility Formation

At the within-person level, the models without and with covariates yielded positive cross-lagged paths: AI perception predicting social presence and social presence predicting

perceived credibility. The findings suggest that when a person began to see comAI more as a social actor, they later felt more as if they were in the presence of another social being, and that this social presence affected perceived credibility within individuals, not just between different people. The longitudinal findings align with our cross-sectional findings and prior experimental research demonstrating that an agent-like conversational presentation mode (vs. tool-like static text) positively affects social presence, thereby increasing credibility perceptions (Anderl, Klein, Sarigül, et al., 2024).

We acknowledge that, particularly for comAI like ChatGPT, experiences at one time point may not simply reflect a carry-over of prior perceptions. Instead, they may also be shaped by interactions occurring between waves. For example, social presence may increase not only because of a social actor perception, but also because of more (relational vs. functional) interactions with the system. To disentangle these mechanisms, future (experimental) studies could include direct measures of interaction style or analyze interaction logs.

Although both the paths from the predictor to the mediator, and from the mediator to the outcome were significant (though very small in size), we did not find a significant total effect of AI perception on perceived credibility. In addition, the indirect effect was only marginally significant and was observed only in the model with covariates, which is why we cannot speak of a longitudinal mediation effect. However, the reverse path from social presence to AI perception was significant, indicating a reciprocal relationship between the two. This exploratory finding suggests that AI perception may not only act as an antecedent but also an outcome shaped by prior interaction experiences. Other reverse paths from credibility, enjoyment, and perceived intelligence to AI perception were not significant.

We did not find significant longitudinal cross-lagged paths of AI perception on enjoyment and perceived intelligence. However, an additional noteworthy, though not entirely surprising, finding was that social presence significantly influenced enjoyment and perceived intelligence (in the model with covariates) over time. This pattern pointed to a potential serial mediation, in which social presence may influence affective and cognitive responses, which in turn contribute to credibility formation. The temporal ordering of effects is consistent with prior literature indicating that social presence may, through enjoyment (Breves, 2023), enhance heuristic processing, which, in turn, leads to more positive evaluations (Jun et al., 2017). Although we could not detect these effects, future research could investigate sequential and potentially reciprocal processes of credibility formation.

Taken together, these findings suggest that the effects underlying credibility formation appear to unfold rather situationally and at the same time point than over the long term. People's perceptions of AI as a tool or a social actor may change quickly depending on the task at hand and the quality of the output (e.g., whether the system hallucinates).

A reason for the absent or small within-person effects in our study might lie in the 2-month distance between the measurement points, resulting in a 4-month lag between the measurement of the predictor and the dependent variable in the mediation paths. This extended time frame, initially employed to capture medium-term processes, may have weakened the situational effects, as they can dissipate when the time lag does not match the underlying temporal dynamics (Cole & Maxwell, 2003). Importantly, we found significant cross-lagged paths from AI perception to social presence, and from social presence to credibility, indicating that these relationships do operate over time. However, they appear

to be rather temporary than cumulative. Accordingly, overall credibility levels remained relatively stable. Neither advances in AI capabilities nor increased user experience over time (e.g., more effective prompting) led to higher credibility perceptions.

Another possible explanation for the largely nonsignificant within-person paths may be the limited variability in participants' comAI perception over time. Across the study period, relatively few participants perceived their favorite AI as a social actor, suggesting that LLM-based chatbots are predominantly used for functional purposes (e.g., as information search, creativity, and productivity tools), rather than for emotional support (Phang et al., 2025). Consequently, few participants switched from tool to social actor (or vice versa) across time points, resulting in limited within-person variation. At the same time, the significant between-person differences (random intercepts) indicated that some users tend to perceive AI as a social actor, whereas others do not. Considering recent scientific discourse about (harmful) synthetic relationships with com AI (Starke et al., 2024), our findings suggest that most users have a stable "tool" framing over time for general-purpose comAI like ChatGPT or Alexa, proposing an interesting avenue for future studies.

Inconsistent Modality Effects (Over Time)

Regarding users' primary interaction modality, we did not systematically find significant effects in the cross-sectional analyses. At T1, higher social presence and enjoyment were observed for text (vs. voice), whereas (only) at T3, perceived credibility was, as expected, higher for voice (vs. text). The latter aligns with prior research, showing that information delivered via voice tends to be perceived as more credible than the same information delivered via text (Anderl, Klein, Sarigül, et al., 2024).

One likely explanation for the scattered effects is the distribution of interaction modalities in our sample. Only about 10% of participants primarily used voice, mostly via Alexa or Apple AI/Siri, while text remained the most common and, for many users, the most familiar, comfortable, and natural modality of engaging with AI. This may partly be because voice interfaces, such as ChatGPT's voice mode, have become available only relatively recently and still may be perceived as somewhat artificial by many users.

In addition, voice- and text-based interactions may be associated with distinct use cases, which may contribute to the heterogeneous effects. Voice-based interaction may be used mainly for short, convenience-oriented tasks (e.g., asking a quick question) or in hands-free contexts (e.g., requesting navigation while driving), whereas text-based interaction may be preferred for more complex and lengthy tasks (e.g., rewriting a text). This perspective may also help explain the reverse effects of modality (text vs. voice) on social presence and enjoyment observed in the present study. Interacting with conversational agents via text has been shown to decrease mental workload (Le Bigot et al., 2004) and increase processing fluency (Zierau et al., 2023), which might make people more comfortable, in turn increasing social presence and enjoyment.

From a methodological viewpoint, unlike prior experimental studies conducted in controlled online study settings that exposed participants to a specific task, the current study focused on users' real-world experiences with their favorite language-based AI

systems. These real-world usage patterns and prior user experiences may have weakened or altered the modality effects typically seen in experimental settings, possibly due to between-person differences in personality traits, communication habits, or accessibility needs.

Importantly, actual modality *use* should be distinguished from underlying preferences. Usage patterns are highly context-dependent, and it is unclear whether changes over time (i.e., from text to voice or vice versa) reflect genuine preference changes or changes in situational constraints or habits. For example, individuals may prefer voice but rely on text in certain contexts (e.g., shared spaces) where speech interaction feels intrusive.

Finally, this study's sample size was too small to test longitudinal modality effects robustly. Future research should use larger samples and (experimental) designs that account for individual differences, task types, and contexts. In particular, increased availability of voice interfaces in LLM-based comAI may reshape usage patterns and perceptions (Mahmood et al., 2025). More research is also needed on users' motivations for combining voice and text modalities.

Practical Implications

Our findings have implications for developers, designers, and policymakers. As organizations increasingly deploy LLM-based chatbots, developers and designers must recognize that some users perceive their favorite comAI as social actors. This tendency is likely to be heightened by anthropomorphic design cues, such as a casual conversational tone or a human-like avatar. However, even if no such cues are intentionally employed, users might view their comAI as a social entity, with which they can exchange ideas, talk, and engage in social interaction, just because of the system's conversational nature. Users may perceive information as more credible because the information source appears human-like, not because it is actually more reliable or accurate (Abercrombie et al., 2023). Considering the flaws and imperfections of these systems in terms of reliability and correctness of answers, practitioners should avoid over-humanizing comAI systems. One measure to avoid manipulating trustworthiness could be to adapt marketing language to a more neutral and objective tone, omitting emotionally charged words, such as "friend," "care," "love," and "empathy."

To protect users from inflated credibility perceptions caused by human-like design features and thereby reducing the risk of users believing and spreading false information, policymakers and organizations must ensure that transparency policies, such as those outlined in Article 50 of the EU AI Act (European Union, 2024), are not only implemented rigorously and responsibly but also further developed. Particularly when comAI systems exhibit highly human-like behavior or appearance, it may be essential that users are regularly reminded that they are interacting with a technological tool, not a human being. This ongoing clarification may reinforce critical awareness, especially when using comAI systems specifically designed for emotional support, social interaction, and relationships, such as Replika or Character.AI.

Limitations and Future Research

Some limitations should be kept in mind when interpreting the results: First, we assessed all variables by asking participants about their subjective perceptions of their favorite comAI system, which is why we cannot rule out biases due to social desirability. Although some participants might treat their favorite AI as a social entity, they might have refrained from selecting the social actor option because they are aware that it is ultimately a technological system. In contrast to our dichotomous AI perception measure, future research could consider a more fine-grained, continuous measure (e.g., using a slider scale) or a multidimensional scale that captures both functional and relational use aspects (e.g., S. Xu & Li, 2024). This would increase variance and, thus, informational value. Moreover, future research could incorporate behavioral measures, for example, by analyzing chat logs, to examine the extent to which people treat comAI as tools or as social actors. In the current study, we investigated the (source) credibility perceptions of comAI systems, rather than specific pieces of information (Anderl, Klein, Sarigül, et al., 2024). However, people's perceptions of a comAI system's credibility may be associated with different actors, such as its human developers or the company behind it (Van der Goot, 2022). Future research could look into these differential source orientations.

Second, as this study was based on data collected for a larger research project, we could not manipulate information accuracy (as did studies with experimental designs on similar research questions; e.g., Anderl, Klein, Sarigül, et al., 2024; Spatharioti et al., 2025). However, given the well-documented error-prone nature of current comAI systems (Fletcher et al., 2025; Kalai et al., 2025), we assumed that individuals are likely to encounter both accurate and inaccurate information while interacting with comAI. It remains an open question, though, what kinds of inaccurate information, including factual errors, logical contradictions, or nonsensical hallucinations (Cossio, 2025), participants encounter and whether these affect users' credibility perceptions in different ways.

Third, we tested our hypotheses using a sample drawn from a Western, educated, industrialized, rich, and democratic (WEIRD) society (Henrich et al., 2010), focusing specifically on active comAI users. This limits the generalizability of our findings. Future research should investigate whether our results can be transferred to different cultural contexts (Folk et al., 2025; Liu et al., 2024). Moreover, it would be valuable to take into account the impact of prior experiences with comAI, for example, by examining differences in credibility formation between long-time users and newcomers.

Conclusion

Based on a four-wave longitudinal survey, we investigated whether active users' credibility perceptions of their favorite comAI depend on whether they perceive it as a tool or a social actor, as well as on their primary usage modality. Cross-sectionally, we demonstrated that the relationship between AI perception and credibility is mediated by users' enjoyment and perceived intelligence of the comAI system, and, in the first wave, by feelings of social presence. RI-CLPMs, both without and with age and gender as control variables, indicated positive cross-lagged effects of AI perception on social presence and of social presence

on perceived credibility, but no longitudinal mediation. Our findings offer novel insights into the psychological processes underlying credibility perceptions of active AI users over time. By focusing on longitudinal field data, our study contributes to the field of Human-Machine Communication by complementing existing experimental findings on the impact of system characteristics on credibility perceptions of comAI.

CRedit Author Statement

Stefanie H. Klein contributed to conceptualization, methodology, data curation, formal analysis, writing the original draft, and visualization; and **Sonja Utz** contributed to conceptualization, methodology, writing review and editing, supervision, and funding acquisition.

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AI/Tool Disclosure

The authors used Microsoft Copilot and OpenAI GPT-5.5 to assist with code debugging and documentation, and to improve the manuscript's language, formatting, and readability.

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Application for Open Science Badges



This article has earned the Center for Open Science badges for Open Data and Open Materials through Open Practices Disclosure. The authors have made their data and materials freely accessible at <https://doi.org/10.17605/OSF.IO/2A7MY>. The article also earned a Preregistered badge for having a preregistered design available at <https://doi.org/10.17605/OSF.IO/F3JYC>.

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